

WASP-36b

R-1070

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_170314 · created 2026-07-30T13:49:18+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457826.6866 +/- 0.0024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.166 +/- 0.05
Transit depth (Rp/Rs)^2	2.77 +/- 1.65 [%]
Orbital Inclination (inc)	88.12 +/- 2.88
Airmass coefficient 1 (a1)	1.41 +/- 0.82
Airmass coefficient 2 (a2)	-0.22 +/- 0.73
Scatter in the residuals of the lightcurve fit is	59.62 %
Best Comparison Star	#1 - [530, 425]
Optimal Aperture	3.0
Optimal Annulus	6.5
Transit Duration (day)	0.0933 +/- 0.0084

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.166 $\pm$ 0.05 vs 0.1368 (deviation 0.4 σ)
Duration fit vs published	0.0933 d vs 0.0758 d (deviation 1.44 σ)
Mid-time O-C	-5.8 min
Depth SNR	2.3
Residual scatter	59.62 %

Light curve

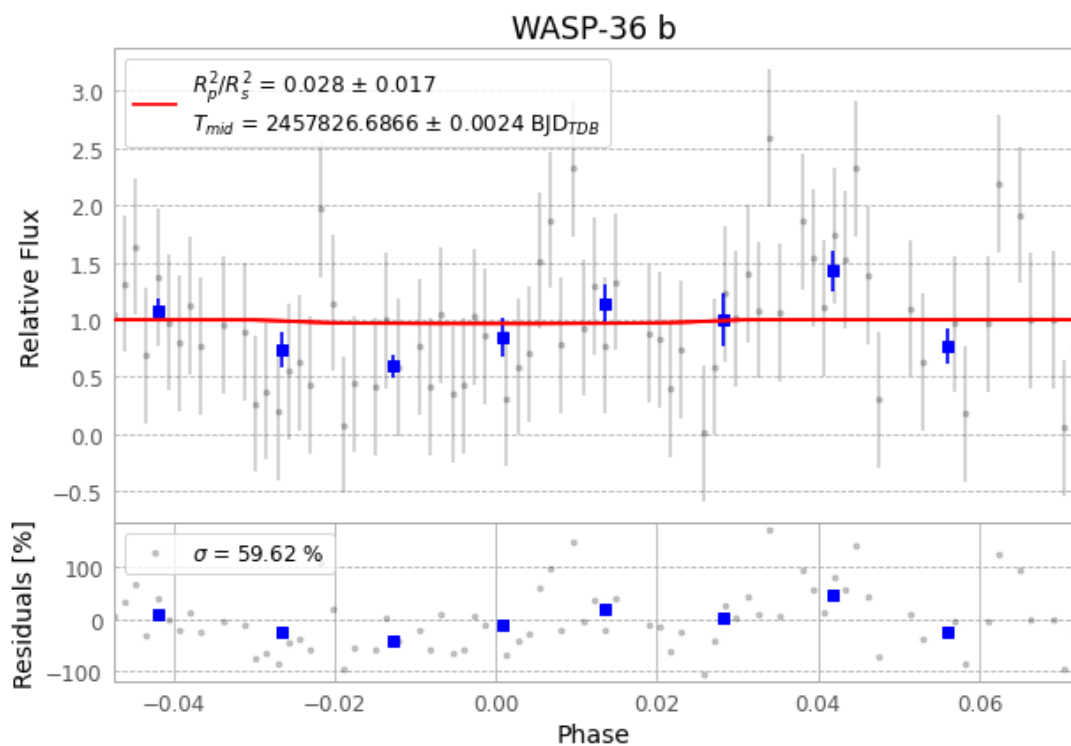


Figure 1 — FinalLightCurve_WASP-36 b_2017-03-13.png (SHA-256 e57208ef67367cad...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [331, 398], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b8d42be92c3e875c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

89 FITS frames (58 MB), WASP-36170314023612.FITS → WASP-36170314070012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-170314.log (82ee981635b5...), FinalLightCurve_WASP-36 b_2017-03-13.png (e57208ef6736...), FinalLightCurve_WASP-36 b_2017-03-13.csv (7661426218ef...)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 13:49Z.